CU-423 RoboLink Sensors

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06-May-2026

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What are we doing today?

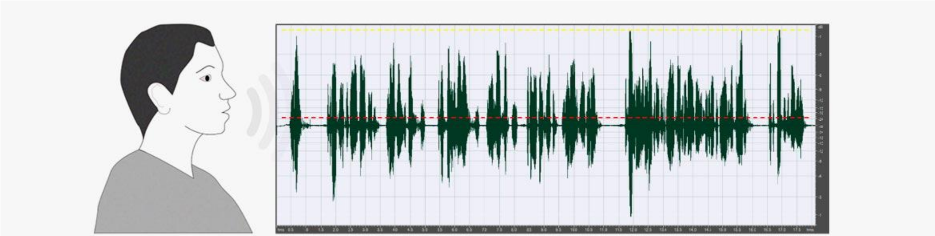
- Data acquisition
  - Analog and digital signals.
- Sensors, what? why? How?
- Basic sensor characteristics
- Sensor examples.
- Sensors Research.

Data Acquisition

- The *process of gathering or sampling the signals that resemble real-world physical phenomenon and converting them into a form suitable for processing.*
  - ***Digital*** *form in case of* ***digital processing***.
- What does “digital” mean? What are the possible types of signals from perspective of independent and dependent variables (e.g., time and voltage)? Actually, ***what does signal mean?***

Data Acquisition – Signals

- What is Signal ?
  - Sequence of states in a channel, or a function of an independent of variable.
  - These states can convey information naturally or can be artificially made to convey information.



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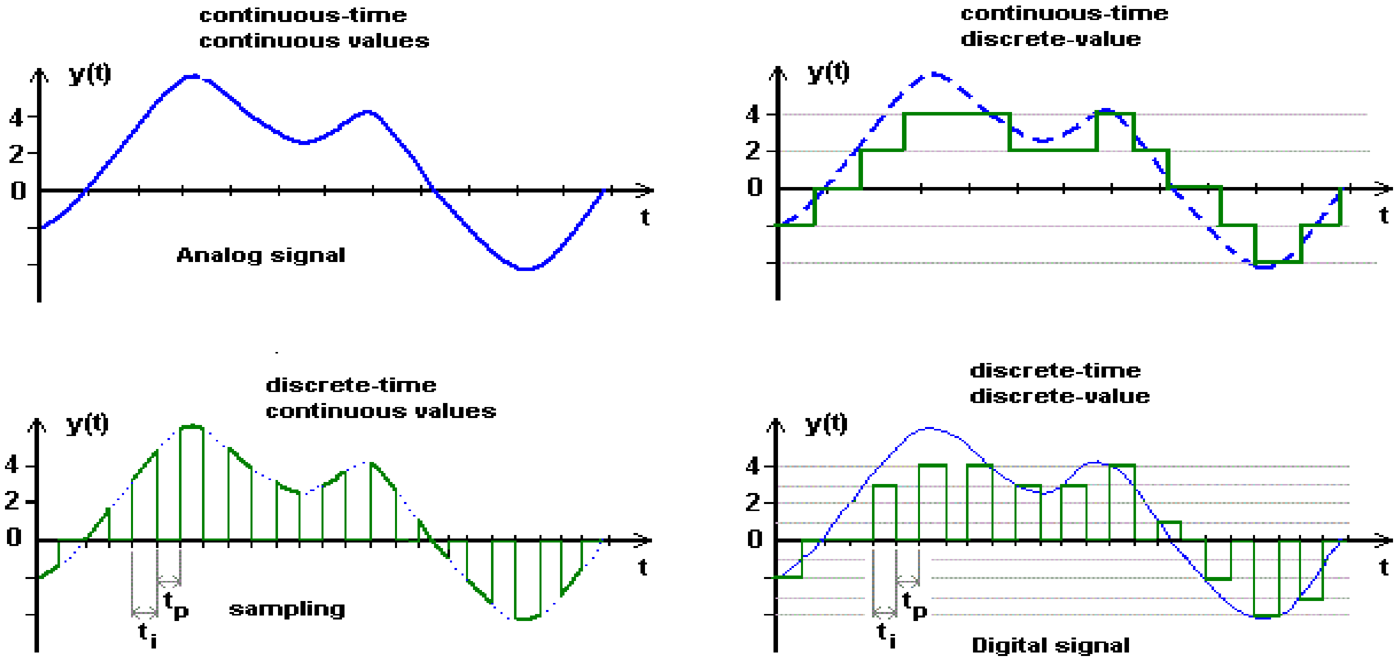
Information as function of time

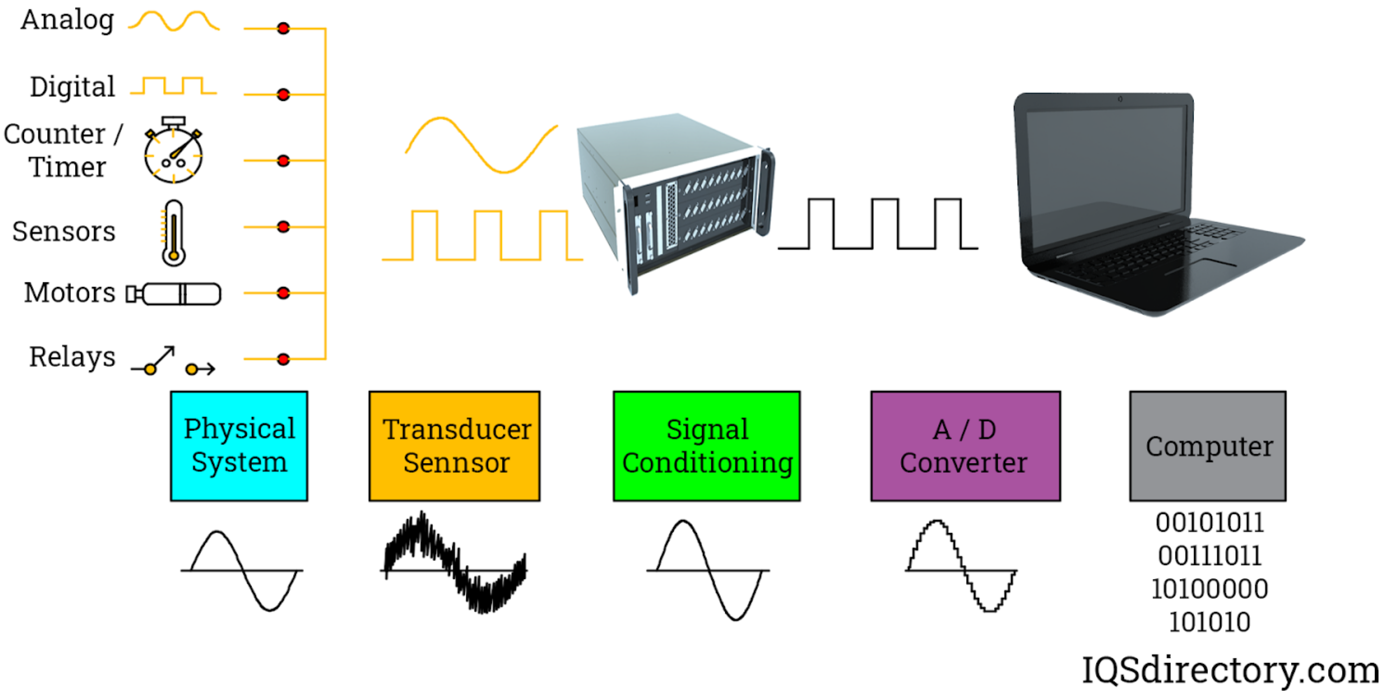
Information as function of space! – spatial signal

Data Acquisition - Signals

- Independent variable (x-axis) (e.g., time): Continuous – Discrete
- Dependent variable (y-axis) (e.g., voltage): Analog - Digital



Data Acquisition – The process



Data Acquisition - Arduino

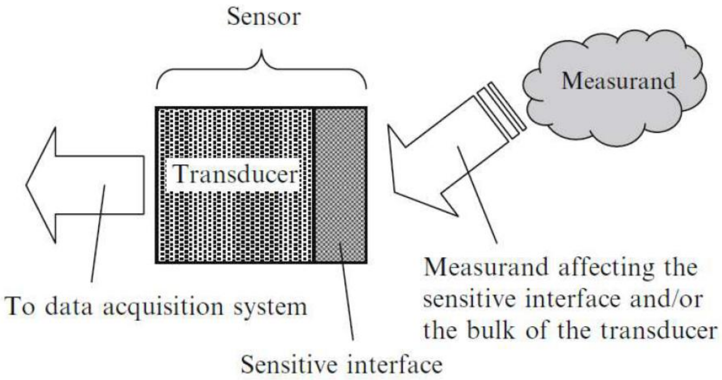
- What are the typical input types through which an Arduino can "acquire" inputs or signals?
  - *Digital IO*
  - *Analog IO*
  - *Communication Protocols (e.g. USB) (digital too).*
- **Regardless of the acquisition method, the data processing happens in digital form.**

Introduction to Sensors

- Sensors:
  - A sensor is a device which responds to stimuli, or an input quality, by generating processable outputs.
  - These outputs are ***functionally*** related to the input stimuli which are generally referred to as measurands.
  - A sensor is commonly made of two major components: a sensitive element and a transducer.

Introduction to Sensors

- Sensitive element has the capability to interact with a target measurand and cause a change in the operation of the transducer.
- Transducer is a device that converts one type of energy to another.



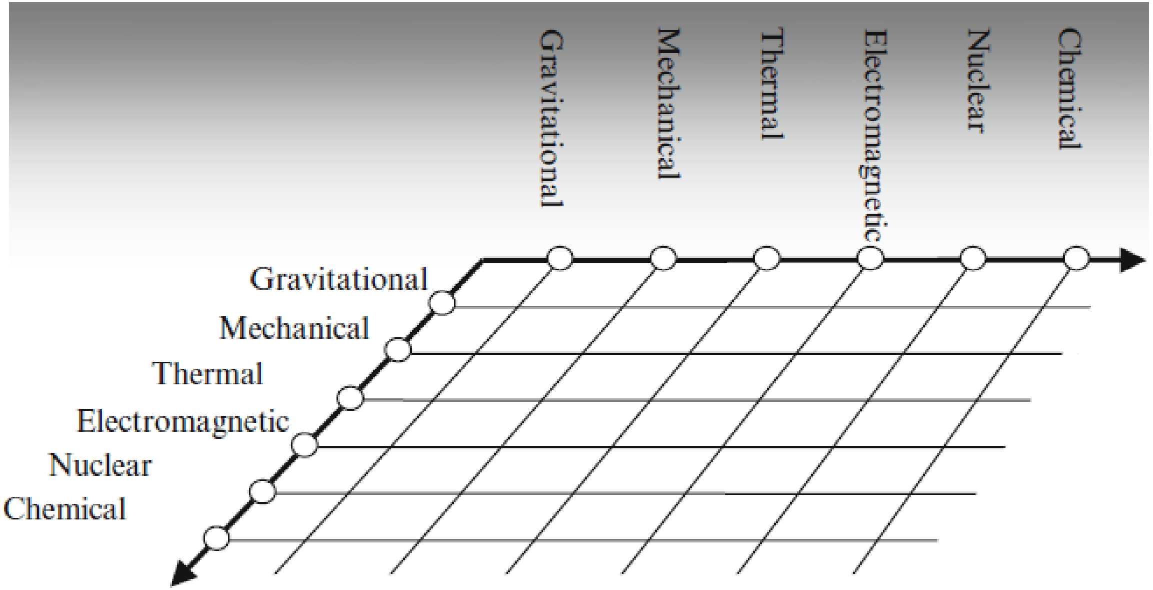
Introduction to Sensors

- Major forms of energy

Type of energy	Occurrence
Gravitational	Gravitational attraction
Mechanical	Motion, displacement, mechanical forces, etc
Thermal	Heat, internal energy (temperature)
Electromagnetic	Electric charge, electric current, magnetism, electromagnetic wave energy
Chemical	Energy released or absorbed during chemical reactions
Nuclear	Binding energy between nuclei—binds the subatomic particles of a matter

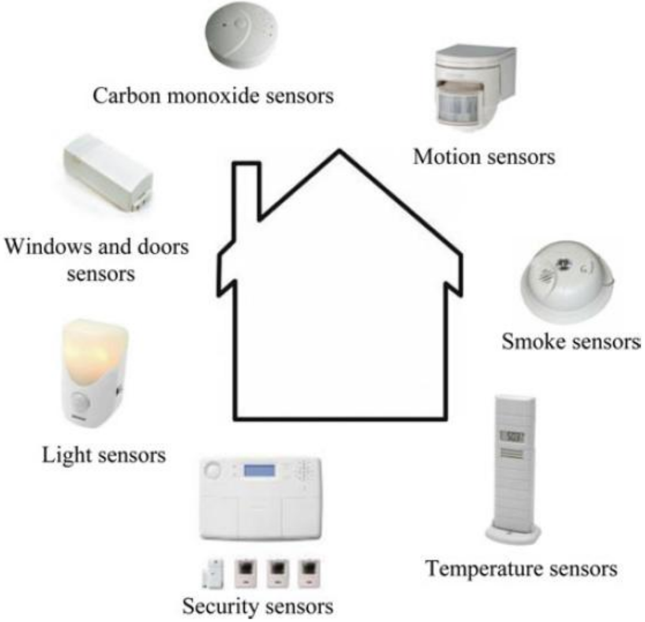
Introduction to Sensors

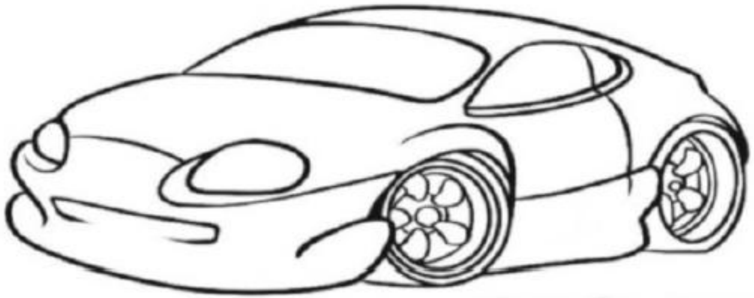
- Transducer changes one form of energy to the other



Introduction to Sensors

- Sensors are everywhere!





Essential drive sensors
Pressure
Mass air flow
Atmospheric pressure
Oxygen
CO_2
Rotational speed
Petrol level
Pedal position
Angular position
Engine temperature
Oil level
Crankshaft position

Safety sensors
Safety distance
Tilt
Torque
Steering wheel angle
Acceleration
Belt

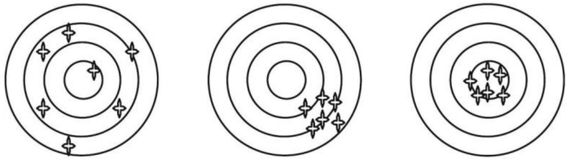
Convenience sensors
Air quality
Humidity
Temperature
Rain
Seat posiion

Sensor Static Characteristics

- Accuracy:
  - Accuracy of a sensing system represents the **correctness of its output** in comparison to the actual value of a measurand.
  - To assess accuracy, either the system is benchmarked against a set of standard inputs, or the output is compared with a measurement system with a superior accuracy.

$$Accuracy = \left(1 - \frac{|error|}{true\ value}\right)$$

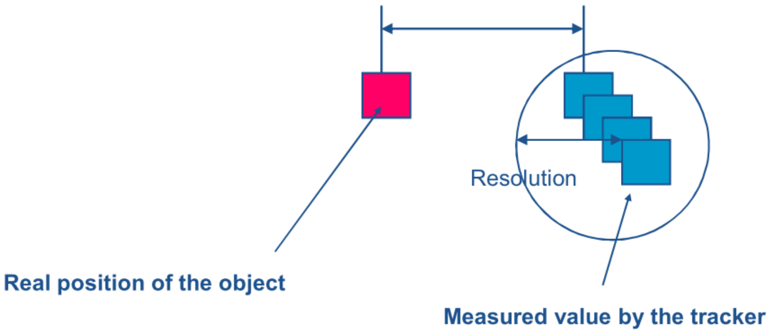
- Precision:
  - Precision represents capacity of a sensing system to **give the same reading** when repetitively measuring the same measurand under the same conditions.
  - The **precision is a statistical parameter** and can be assessed by the standard deviation (or variance) of a set of readings of the system for similar inputs.



Low precision - low accuracy | High precision - low accuracy | High precision - high accuracy

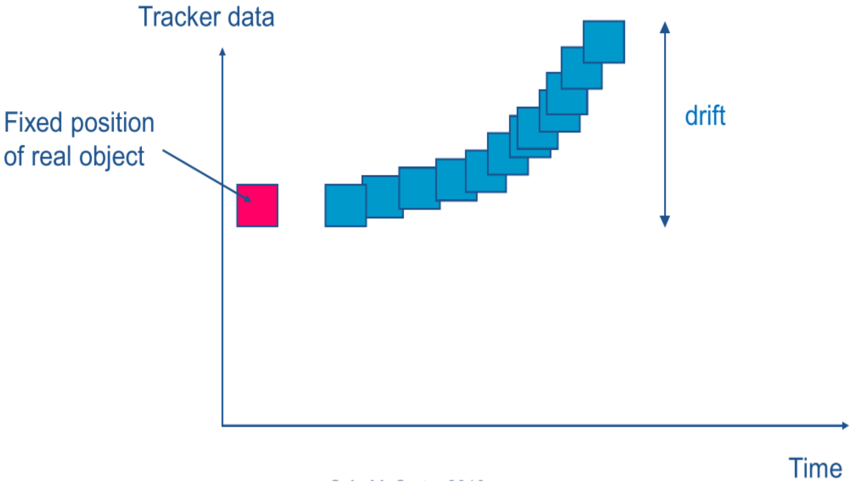
Sensor Static Characteristics

- Resolution
  - Resolution (or discrimination) is the ***smallest amount of change in the input that can produce a detectable increment in the output***.
  - Resolution depends solely on the sensor hardware, while accuracy depends on sensor hardware and the measurement environment
  - Resolution is strongly limited by any noise in the signal.



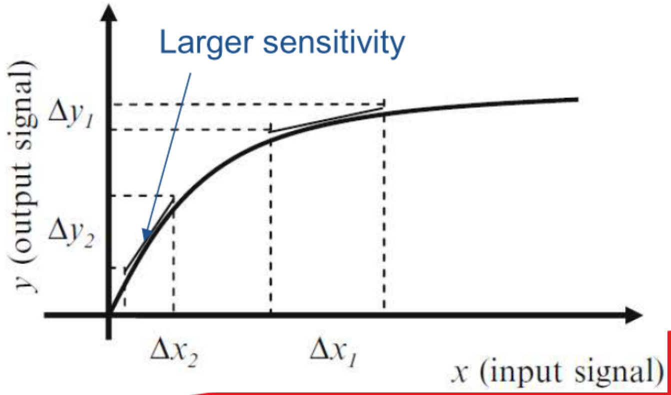
Sensor Static Characteristics

- Drift
  - Drift is observed when a *gradual change in the sensing system's output* is seen, while the measurand actually remains constant.
  - Drift is the undesired change that is unrelated to the measurand.
  - It is considered a systematic error, which can be attributed to interfering parameters such as and the sensor's *materials degradation*.



Sensor Static Characteristics

- Sensitivity
  - The ratio between the incremental change in the sensor's output (Δy) to the incremental change of the measurand in input (Δx). I.e., the slope of the calibration curve, $y = f(x)$,.
  - Ideal sensor has large and constant sensitivity in its operating range.
  - Thermometer: 1V change for 0.1°C change -> Sensitivity = 10V/°C

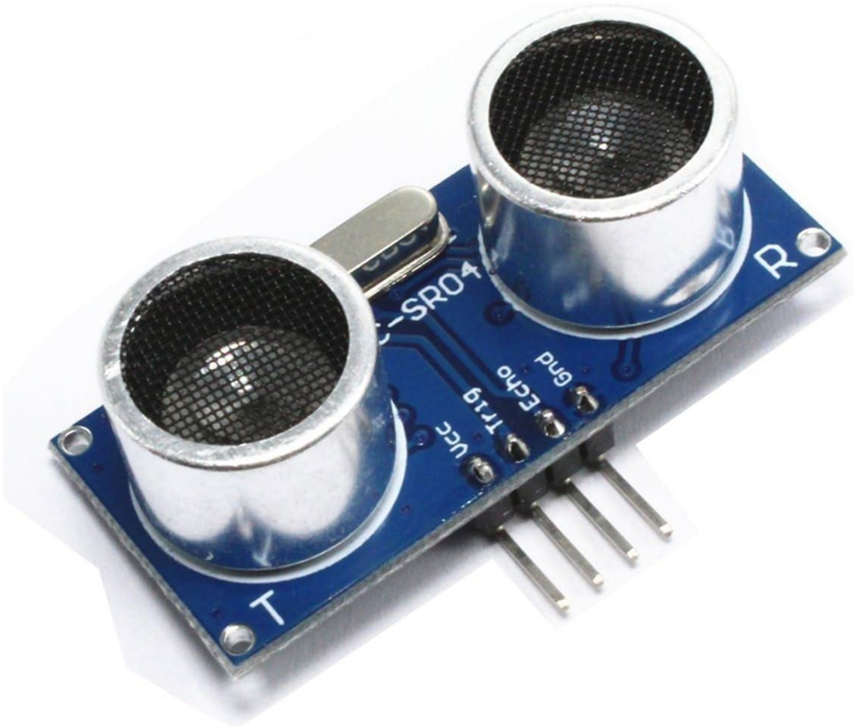


Sensors Characteristics

- Sensors Qualities:

  - Reliable
  - Accurate
  - Low Cost
  - Low Power consumption
  - Should not affect the measurand
  - Life time
  - Sensitive to measurand
  - Insensitive to noise / interference
  - Environmentally friendly
  - Documentation / Datasheet
  - Response behavior
  - Selectivity

The Ultrasonic Distance Sensor

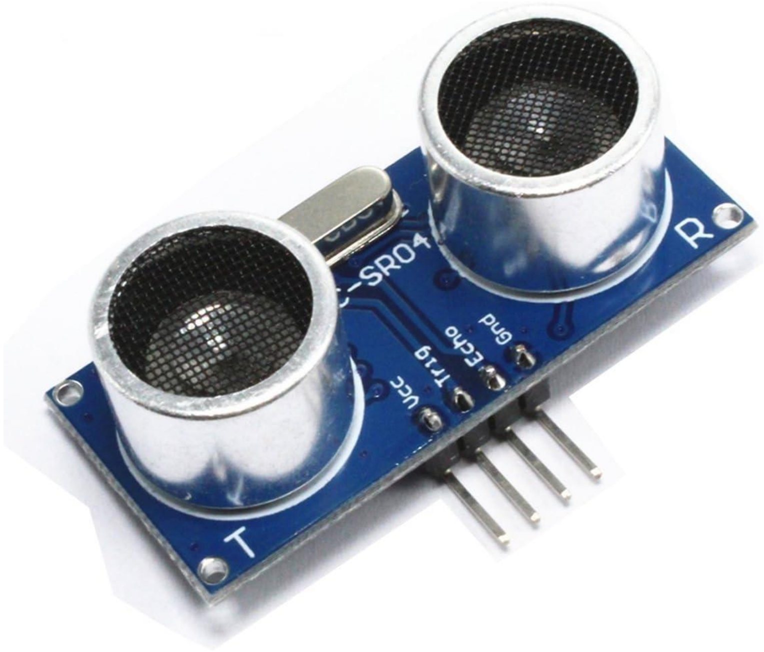


The Ultrasonic Distance Sensor

- Used to measure distance to obstacles
- Physical principles:
  - Sound waves have known speed
  - Sound waves reflect from surfaces
  - Distance is related to speed and time

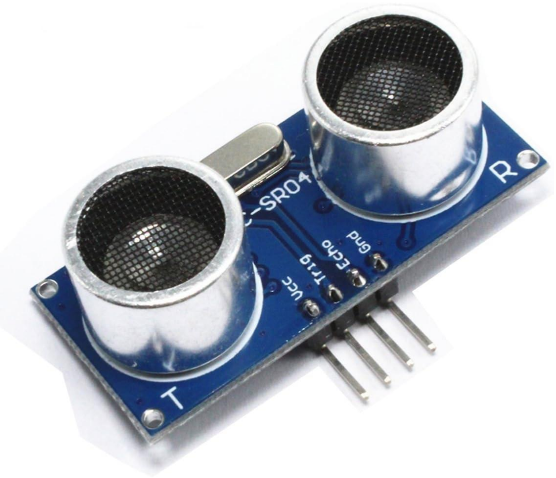
$$v(\text{Speed}) = \frac{x(\text{Distance})}{t(\text{Time})}$$

$$x = vt$$



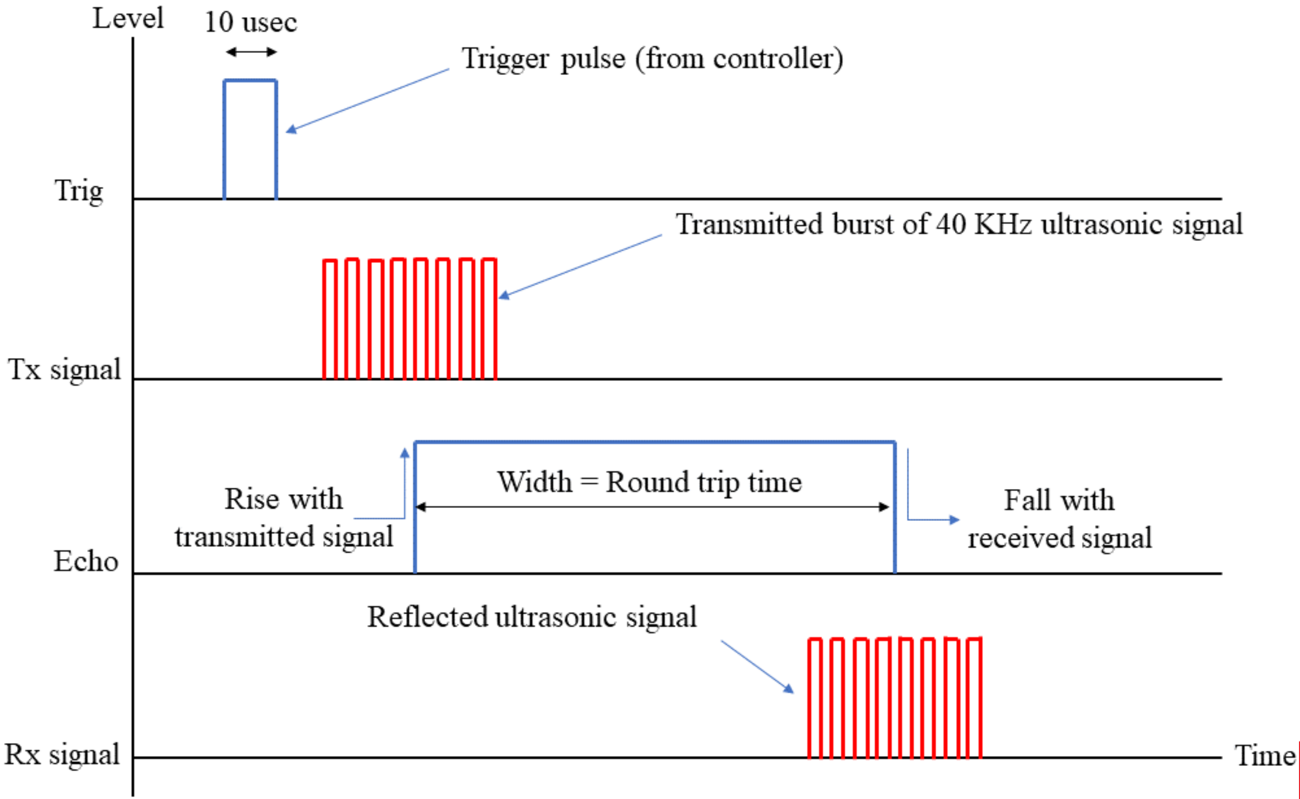
The Ultrasonic Distance Sensor

- Working Principle
  - On a “Trigger” signal, the sensor outputs an ultrasonic (not hearable) sound wave pules
  - The sensor waits for the reflected wave.
  - When the “Echo” comes back, the sensor issues an electrical signal.
  - By measuring the time between the trigger and echo signal, the distance can be calculated.



The Ultrasonic Distance Sensor

- Signal Diagram



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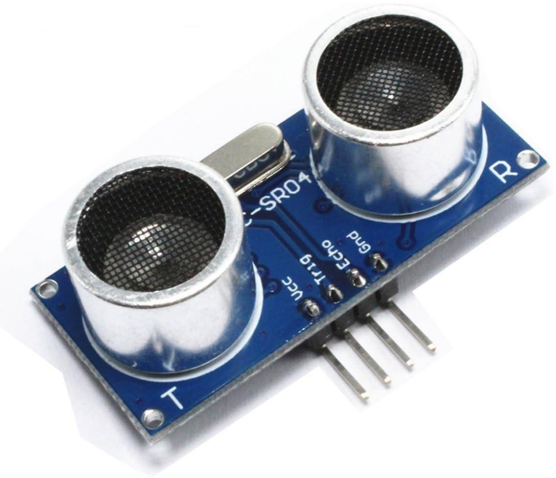
The Ultrasonic Distance Sensor



- Equation for calculating the distance D in meters

$$D = 343 \times \frac{T}{2}$$

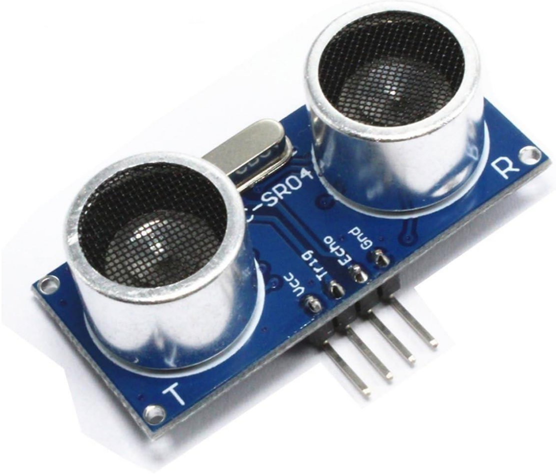
- Speed of sound: 343 m/sec, T: time in seconds
- Why the “/2”?
- If the time is given in microseconds and distance in cm

$$D = 343 \times \frac{T}{2} \times \frac{1}{1{,}000{,}000} \times 100 = 343 \times \frac{T}{20{,}000}$$

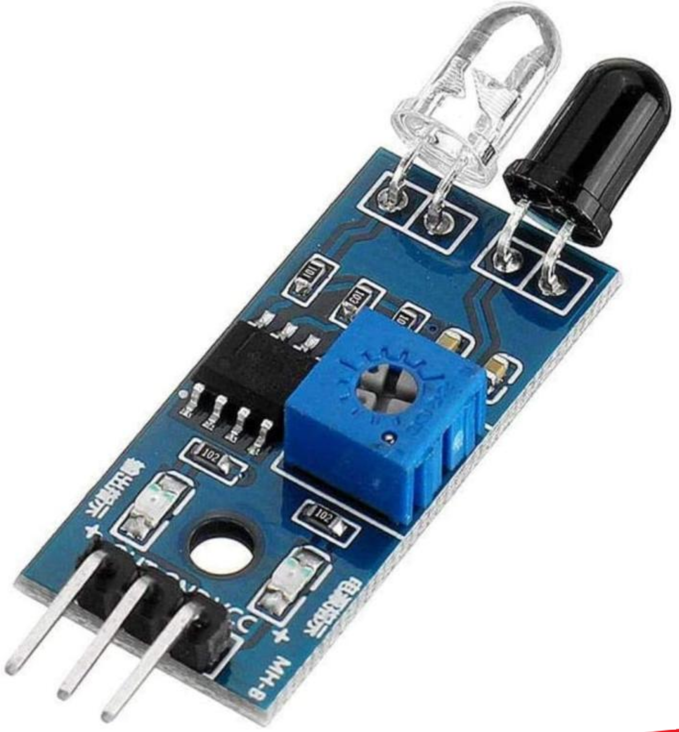
The Ultrasonic Distance Sensor

- Specifications

Operating Voltage	**3.3V to 5V**
Acoustic frequency	40 KHz
Measurement Angle	15°
Range (claimed)	5.5 m
Trigger format	20 usec pulse.
Output type	Pulse width modulated output
Working temperature	-10 to +60 ℃

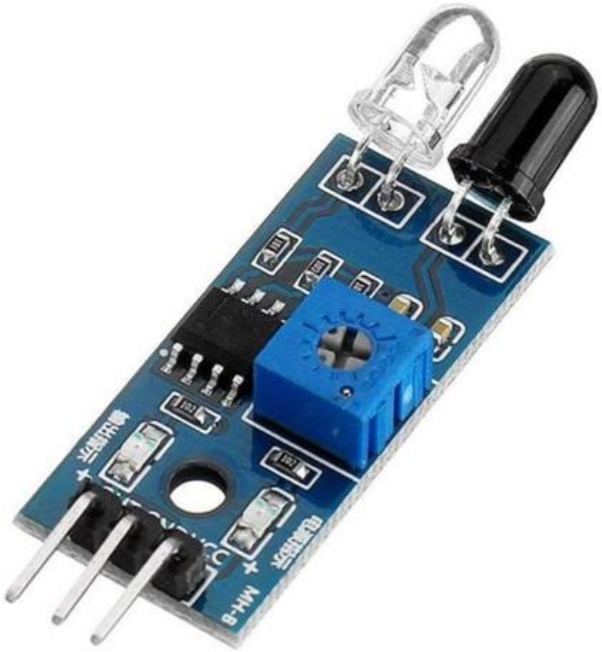


The IR Obstacle Avoidance Sensor



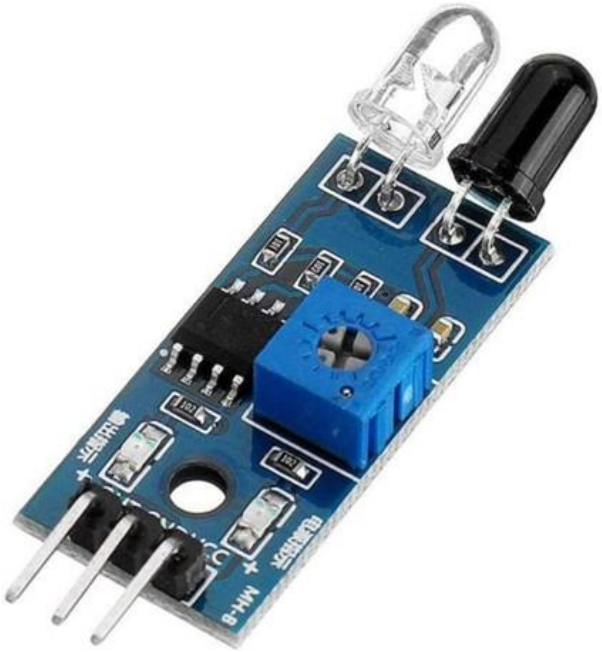
The IR Obstacle Avoidance Sensor

- Used to detect obstacles
- Physical principles:
  - Electromagnetic waves reflect from surfaces.
  - The detection of a reflected IR wave indicates the existence of an obstacle



The IR Obstacle Avoidance Sensor

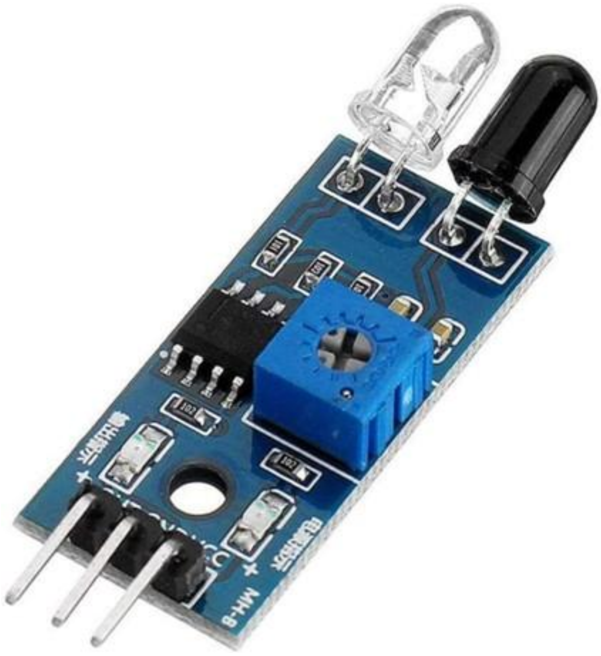
- Working Principle
  - The sensor sends IR pulses continuously
  - The received signal is filtered to detect only the transmitted IR frequency
  - Once the signal is received, a voltage is asserted on the output pin, which can be an input to the Arduino



The IR Obstacle Avoidance Sensor

- Specifications

Operating Voltage	**3.3V to 5V**
Output Type	On/Off
Distance Range	2 to 40 cm
Working Temperature	-10° C to +50° C
I/O Interface	GND, VCC, OUT
Dimensions	45×16×10 mm
Weight	9g



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Research Time!

Sensors Research

Group	Sensor
Group 1	Distance
Group 2	Light intensity
Group 3	Temperature
Group 4	Touch sensing
Group 5	Current
Group 6	Humidity
Group 7	Sound (noise)

Sensors Research

- What your presentation should include:
  - A picture of the sensor
  - Working principle: how does it work?
  - Specifications
  - Interface: Can we connect it to our Arduino? How?
  - Research time: 30 mins
  - Presentation time: 5 mins

Group	Sensor
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Specifications Example

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Break!

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Questions?

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